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Palaeomagnetism and $^{36}Ar/^{40}Ar$ vs. $^{39}Ar/^{40}Ar$ isotope correlation ages of dyke swarms in central Kerala, India: Tectonic implications

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Abstract

Palaeomagnetic results and whole-rock $^{36}Ar/^{40}Ar$ vs. $^{39}Ar/^{40}Ar$ isotope correlation ages are presented for dyke swarms occurring within the granulite terrain in central Kerala (west coast of India). Palaeomagnetic data show that the NNW–SSE leucogabbro dykes sampled are normally magnetised ($D = 307$ and $I = -57$), whereas most NW–SE dolerite dykes are reversely magnetised ($D = 163$ and $I = 61$). One dolerite dyke shows normal polarity. The present $^{40}Ar/^{39}Ar$ isotope ages for dolerite dykes and previously reported K-Ar ages for leucogabbros suggest 81 ± 2 Ma for acquisition of the leucogabbro directions and 69 ± 1 Ma for acquisition of the reverse magnetic directions within the dolerites.

Based on palaeomagnetic and isotopic age data, the dolerite dykes appears to represent feeder systems for the Deccan volcanic sequences. We suggest that the Deccan eruptions were widespread, and probably extended southward at least into central Kerala. The entire magmatic episode may have occurred between ca. 70 and 65 Ma (during the 31R period probably corresponding to the Cretaceous/Tertiary boundary). The leucogabbros may correspond to the 34N superchron and represent a plume-related magmatic episode precursory to the main volcanic event. These data suggest that an early intraplate plume tectonic setting (since ca. 80 Ma) preceded lithospheric thinning, asthenospheric upwelling and consequent massive flood volcanism during the period 70–65 Ma. This culminated in ocean-floor spreading between the Seychelles–Mascarene and continental India.

1. Introduction

Several Phanerozoic dyke swarms are exposed along the west coast of southern India. These have generally been considered to be related to the Deccan basalts. However, neither reliable palaeomagnetic directions nor precise radiometric ages have been available for these dykes. Similarly, although many conventional K-Ar isotope ages have been reported for Deccan rocks, the duration of Deccan volcanism has been uncertain [1], and relationships between the dykes and Deccan volcanism have not been properly evaluated. Reported K-Ar ages suggest that Phanerozoic mafic dyke intrusions in southern India were episodic [2] and some ages markedly differ from Deccan ages.

The Mascarene plateau, Seychelles bank and

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the Madagascar ridge within the Indian Ocean are composed of discrete Phanerozoic volcanic sequences. These tectonic elements appear to have been contiguous with western continental India in most late Phanerozoic Gondwana reconstructions and the volcanic sequences have been interpreted to represent magmatic events associated with rifting. This implies that some mafic dykes exposed along the western continental margin of India could possibly have been related to these volcanic sequences and not solely to Deccan volcanism. However, only limited data are available for these igneous units occurring over this entire terrain. Systematic palaeomagnetic and $^{40}Ar/^{39}Ar$ isotope studies from the Deccan trap stratigraphic sequences have only recently been

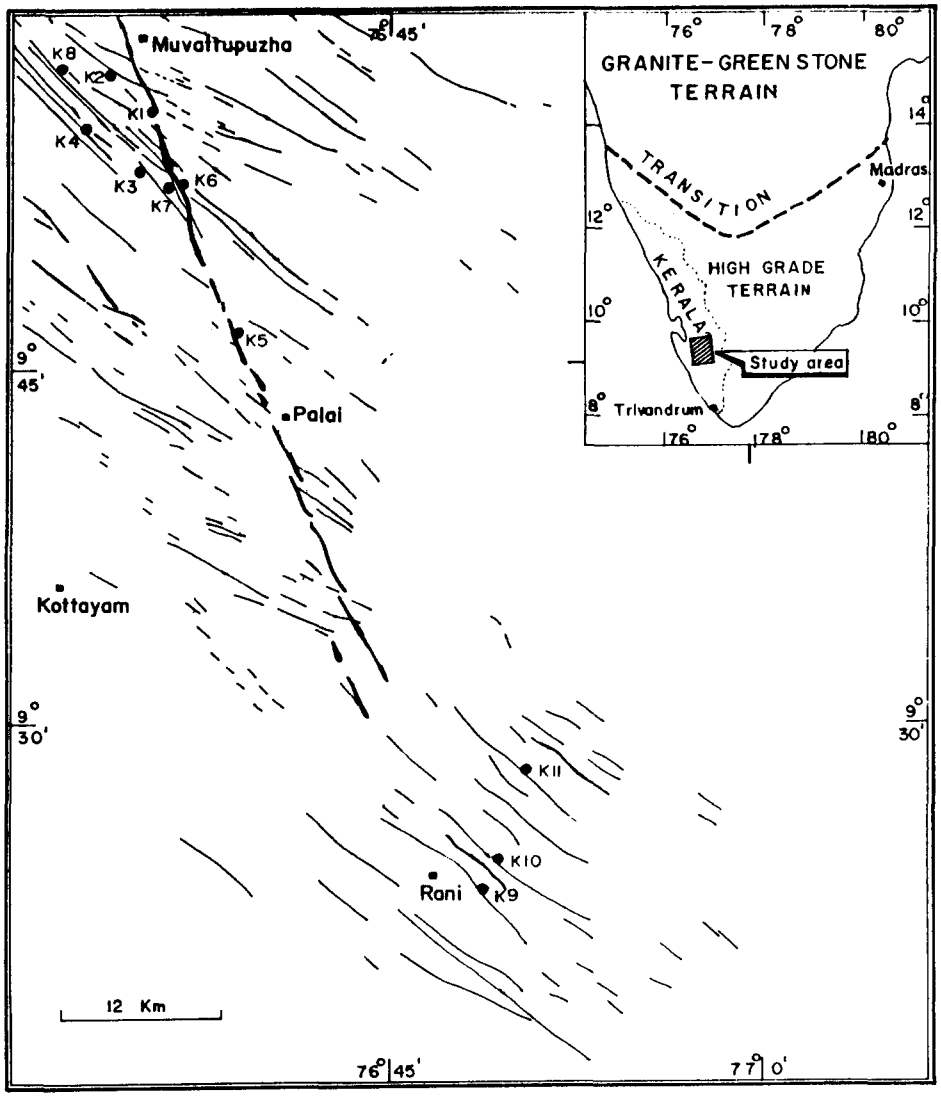


Fig. 1. Geological map of the central Kerala dykes showing the palaeomagnetic sampling sites. The country rock is predominantly charnockite with minor and subordinate high-grade rocks.

available from limited areas. Herein we present palaeomagnetic directions and whole-rock $^{40}Ar/^{39}Ar$ ages for several dyke systems exposed in central Kerala and correlate them with the Deccan sequences.

Deccan flood basalts are regarded as the magmatic responses to the Reunion mantle plume. The Reunion hot spot manifests the clear effects of a mantle plume, and was centrally situated beneath the volcanic outpourings during the eruptions [3,4,5]. Unlike some mantle plumes (e.g., that responsible for the Siberian traps), the ascent of the Reunion plume was linked to initiation of ocean spreading and rifting of the Seychelles platform from India following the Deccan eruptions. However, the cause and effect relationships between volcanism and rifting are unclear [4]. The new data presented herein have interesting implications for these relationships, and the entire tectonic evolution of the northwestern Indian Ocean.

2. Geological setting

The mafic dyke systems examined in this study are exposed between 9° and 10°N along the western coast of India. A generalised geological map with palaeomagnetic site locations is shown in Fig. 1. Country rocks for the dykes include granulite-facies charnockite and subordinate retrograded garnetiferous gneiss. The available geochronological and structural information suggests that the final phase of the granulite tectonothermal activity occurred in the early Proterozoic. The dykes follow predominant fractures that represent extensional features. On the basis of emplacement direction and physical characteristics, the central Kerala dykes may be assigned to two distinctive systems. One includes a NNW–SSE trending leucogabbro dyke (Fig. 1), and the other includes NW–SE trending, medium- to fine-grained dolerites. None of these dykes penetrate into the Tertiary coastal sedimentary rocks. Because they are concealed under the sedimentary deposits the NW–SE dolerites only appear to be truncated towards the coast.

The gabbro dyke is aligned to nearly linear, with only limited offsets along a total strike length of > 100 km. The dolerite dykes vary in length and can reach ca. 10 km, and they parallel each other within a zone that is ca. > 50 km wide. The gabbro dyke is relatively thick (20–50 m), whereas the dolerites are thin, with maximum widths of ca. < 5 m. The dolerites locally crosscut the NNW–SSE trending leucogabbro dyke, proving that the former are younger. The contacts of both dyke systems are sharp, with well-developed chilled margins and steep dips.

3. Petrographic and geochemical characteristics

The petrological and geochemical characteristics of the studied dykes have been discussed in detail elsewhere [6]. Both the dolerites and the gabbro consist dominantly of plagioclase, clinopyroxene and opaque minerals. The dolerites locally contain minor olivine. Apatite is a common accessory mineral and quartz occurs frequently as an interstitial constituent in the gabbro. Plagioclase occurs locally as phenocrysts or as glomerocrysts in an ophitic matrix of plagioclase and pyroxene in the dolerites. In the gabbro the plagioclase is equigranular and clinopyroxene commonly occurs as an interstitial constituent. Micropegmatitic intergrowths of quartzofeldspathic material occasionally occur along some plagioclase grain boundaries in both gabbro and dolerite. In the leucogabbro, slight deuteric alteration of clinopyroxene into pale green amphibole (actinolite) is common along internal cleavages and grain boundaries. However, the amphibole does not occur as independent grains or entirely replace original minerals. Fe-Ti oxides occur as dispersed grains, locally with skeletal shapes.

Geochemical characteristics suggest that the dolerites and the gabbro were derived from discrete parental magmas [6]. In terms of incompatible trace elements, the dolerites are correlative with tholeiitic units within upper stratigraphic units in the Deccan basalts. The gabbros show general geochemical correlations with alkaline-type rocks exposed in northern parts of the Deccan province [6].

4. Palaeomagnetism

A total of eleven sites were sampled for palaeomagnetic investigation across the total geographic distribution of central Kerala dyke swarms. These included eight sites within NW–SE trending dolerites and three within the NNW–SSE trending leucogabbro dyke. At each site at least six oriented block samples were collected within an individual dyke, using a magnetic compass for orientation. These were later set in the laboratory to obtain oriented cylindrical specimens (1″ × 1″) for laboratory measurements.

A Molspin Minispin spinner magnetometer (UK) was used for remanent vector measurements. Specimens were treated by both alternating field (AF) and thermal demagnetisation techniques using a Molspinn AC shielded demagnetiser and an MMTD-1 thermal demagnetiser (Magnetic Measurements, UK) in a field-free space. AC demagnetisations were performed in progressive steps at close intervals (2.5 mT) and thermal demagnetisations on specimens representative of each site up to > 600°C in steps of 50°C. Standard vector plots, stereographic plots and principal component analysis were used to recover the characteristic primary magnetisations.

Sample mean vectors of natural remanence at the NRM level for the central Kerala dykes display a large scatter both within and between sites. Responses of typical specimens to demagnetisations may be seen in the vector diagrams in Figs. 2 and 3. For ten sites the single characteristic magnetic directions could be successfully isolated, after removal of soft (viscous) remanent magnetisation in fields between 10 and 25 mT. The directions thus obtained display good grouping within each site (Fig. 4). Thermal demagnetisations were not successful in isolating characteristic magnetic vectors. The demagnetisations suggest that titanomagnetite is the main magnetic carrier. At one site (K10), the directions are random and a stable magnetic vector could be recovered from none of the samples. At site K4, stable vectors could be achieved at < 20 mT in all specimens, and a good grouping is defined (Fig. 4). However, it can be seen that the magnetisation direction ($D = 242$, $I = 44$) is markedly

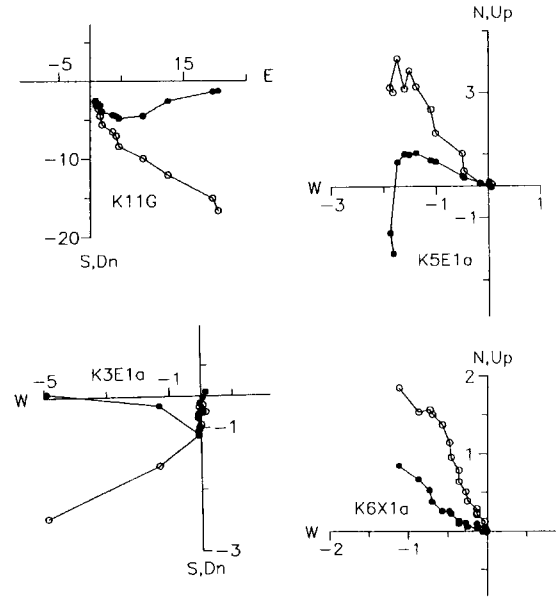


Fig. 2. Zijderveld vector projection diagrams of AC demagnetisations on representative site samples. The units on the X and Y axes are in A/m.

distinct from other site directions of this swarm. Therefore, these anomalous results from this site were not included in the statistical analysis of the palaeomagnetic data of this swarm.

NRM intensities in the gabbro are always within the 1–6 A/m range, whereas the dolerites show a general range of 1–8 A/m. The dolerites rarely exceed this range, although a highest value of 43 A/m was recorded in one sample from site K3. For most samples the intensities fall sharply to < 10% of the original value in the initial few steps (< 10 mT) of demagnetisation. Only a few samples show high coercivity.

The mean stable vectors of the sites are compiled in Table 1. The three sites from the leucogabbro yielded normal polarity with a very low 95% confidence of 12. In the dolerite swarm, apart from the unstable site, six out of seven sites have a reverse magnetisation. In this respect, the dolerite dykes of central Kerala appear generally correlative with the Deccan traps where most volcanic activity appears to have occurred during reverse magnetic polarity [7,8,9].

5. Ages of the dyke intrusions

5.1. Previous K-Ar work

Five whole-rock samples from each of the dolerite and gabbro dyke groups were dated by the conventional K-Ar method (Table 2, [2]). All gabbro samples except for one yielded remarkably consistent ages, at 81 ± 3 Ma. One sample recorded a 62 ± 1 Ma age which may be due to loss of Ar. The K-Ar ages of the dolerite samples range between 50 and 100 Ma and the K-Ar system was disturbed in many samples. A $^{40}Ar/^{39}Ar$ study was therefore undertaken on the dolerite samples.

5.2. New $^{40}Ar/^{39}Ar$ analysis

In an attempt to better constrain the ages of the dyke intrusions, four whole-rock dolerite dyke samples were analysed with $^{40}Ar/^{39}Ar$ incremental release dating methods. The techniques used during the $^{40}Ar/^{39}Ar$ analyses generally followed those described in detailed by Dallmeyer and Ibarguchi [10]: The whole-rock samples were prepared by crushing and sieving to the 80/100 mesh followed by thorough washing. The whole-rock powders were wrapped in aluminum-foil packets, encapsulated in sealed quartz vials, and irradiated for 40 h at the TRIGA reactor at the U.S. Geological Survey in Denver, Colorado. Variations in neutron flux along with the length of the irradiation assembly were monitored with several mineral standards, including MMhb-1 [11]. The samples were incrementally heated until fusion in a double-vacuum, resistance-heated furnace. Measured isotopic ratios were corrected for total system blanks and the effects of mass discrimination. Interfering isotopes produced during irradiation were corrected using factors reported by Dalrymple et al. [12] for the TRIGA reactor. Apparent $^{40}Ar/^{39}Ar$ ages were calculated from corrected isotopic ratios using the decay constants and isotopic abundance ratios listed by Steiger and Jager [13] following the methods described in Dallmeyer and Keppie [14].

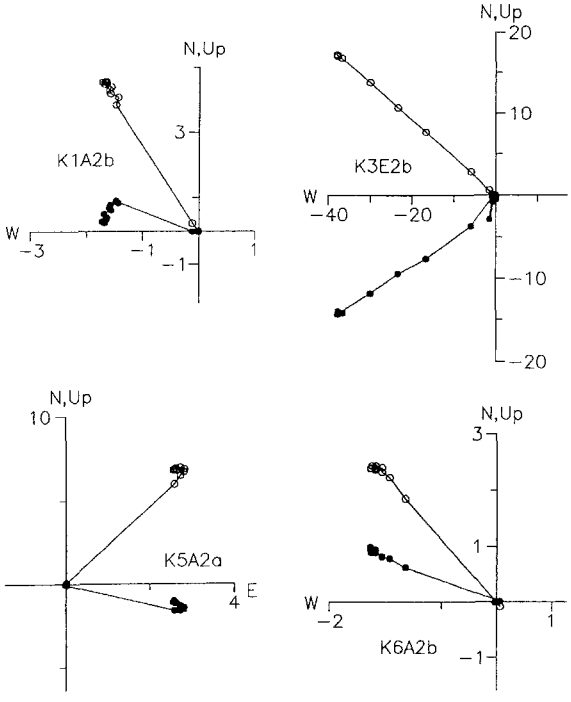


Fig. 3. Zijderveld vector projection diagrams of thermal demagnetisations on representative site samples. The units on the X and Y axes are in A/m.

Intralaboratory uncertainties were calculated by statistical propagation of uncertainties associated with measurement of each isotopic ratio (at two standard deviations of the mean) through the age equation. Interlaboratory uncertainties are ± 1.25–1.5% of the quoted age. Total-gas ages were computed for each sample by appropriate weighting of the age and percentage ^{39}Ar released within each temperature increment. A plateau is considered as being defined if the ages recorded by two or more contiguous gas fractions (with similar apparent K/Ca ratios) each representing $> 4\%$ of the total ^{39}Ar evolved (and together constituting $> 50\%$ of the total quantity of ^{39}Ar evolved) are mutually similar within a $\pm 1\%$ intralaboratory uncertainty. Analysis of the MMhb-1 monitor indicates that apparent K/Ca ratios may be calculated through the relationship $0.518(\pm 0.005) \times (^{39}Ar/^{37}Ar)_{corrected}$. Plateau portions of the analyses were plotted on $^{36}Ar/^{40}Ar$ vs. $^{39}Ar/^{40}Ar$ isotope correlation diagrams. Regression techniques followed the methods of York [15]. A mean square of the weighted deviates

(MSWD) was used to evaluate isotopic correlations.

5.3. Results

The $^{40}Ar/^{39}Ar$ analytical data are listed in Tables 2 and 3 and are portrayed as age spectra in Fig. 5. Samples 3B and 3C were collected within the same dyke whereas samples 2G and 4D were collected at two different sites within other dolerite dykes. The whole-rock samples display internally variable apparent K/Ca spectra, reflecting their polymineralic nature. The four samples are characterised by variably discordant age spectra. Samples 2G and 3B display the least internal age variation, with most intermediate-temperature increments recording similar intrasample ages which define plateaus of 78.7 ± 4.4 Ma (2G) and 71.9 ± 1.5 Ma (3B). $^{36}Ar/^{40}Ar$ vs. $^{39}Ar/^{40}Ar$ isotope correlations of the plateau data are well defined (MSWD < 2.5), and result in inverse ordinate intercepts of 321.7 ± 7.3 and 298.8 ± 2.8. These are not significantly larger than the $^{40}Ar/^{36}Ar$ ratio in the present-day atmosphere, and suggest little contamination with extraneous ('excess') argon components. Using the inverse abscissa intercepts ($^{40}Ar/^{39}Ar$ ratio) in the age equation yields ages of 59.7 ± 5.3 Ma and 69.4 ± 1.1 Ma. Because calculation of isotope correlation ages does not require the assumption of an $^{40}Ar/^{36}Ar$ ratio they are considered more reliable than plateau ages calculated directly from the analytical data. The isotope correlation ages are considered geologically significant, and are interpreted to closely date dyke emplacement.

Whole-rock dyke samples 3C and 4D display markedly discordant apparent age spectra which define total-gas ages of 107.6 ± 4.2 Ma (3C) and 325.6 ± 4.5 Ma (4D). Intermediate-temperature increments evolved from sample 3C (680–1040°C) yielded a well-defined isotope correlation with an inverse ordinate intercept of 357.5 ± 3.0. The inverse abscissa intercept yields an isotope correlation age of 69.9 ± 2 Ma, which is considered geologically significant. The internal apparent age variation displayed by sample 3C is interpreted to reflect the effects of experimental evolution of extraneous argon with a constant isotopic composition. No meaningful isotope correlations are defined by any combinations of incremental data from sample 4D. The ca. 325 Ma total-gas age is considered to have no geological significance and results from contamination with extraneous argon components of variable isotopic composition.

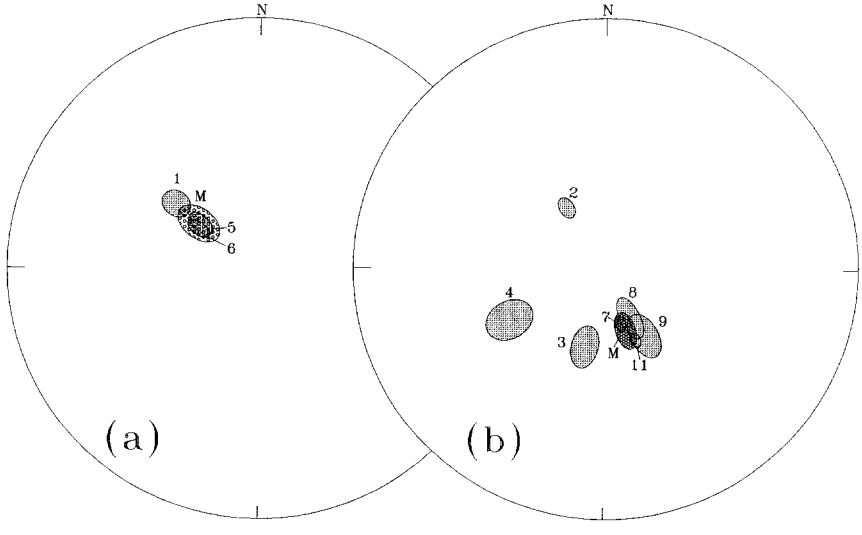


Fig. 4. Equal-area projections of magnetic directions from the samples of the dyke sites. α_{95} ellipses of the characteristic remanent magnetic directions from each site after AC demagnetisations are shown for the gabbro (a) and dolerite (b) sites.

6. Central Kerala dykes and Deccan flood volcanism

Palaeomagnetic results for the central Kerala dolerite dykes yielded a reverse polarity, with the exception of site K2. The dykes have yielded a mean magnetic pole ($\lambda = 34.6$, $\phi = 274.0$) with α_{95} of 8°. In the absence of a reliable APWP for India, the palaeomagnetic poles of the Deccan have been transferred to the North American reference frame to assign palaeomagnetic ages that may be compared to the North American APWP [e.g., 17]. However, recently a synthetic APWP for India that is constrained by data reported from the Eurasian, North American and African plates transferred across the Atlantic and Indian oceans has been reported [18]. This new synthetic APWP is more relevant for comparisons because more data have been integrated [19]. The mean pole of the central Kerala dolerites is compared to the synthetic APWP of India in Fig. 6. It is situated between the 60 Ma mean (50–70 Ma window) and the 70 Ma mean (60–80 Ma window) poles; its α_{95} circle intersects that of the two mean ages. Using the weighted ages in the 50–70 Ma and 70–90 Ma windows [18], the interpreted age for central Kerala can be 69.4 ± 7.8 Ma. $^{40}Ar/^{39}Ar$ isotope correlation ages from two palaeomagnetic samples (table 3) with reverse vectors are 69 ± 1 Ma. There is good agreement between the interpreted palaeomagnetic ages and the $^{40}Ar/^{39}Ar$ ages.

Rao et al. [20] correlated the lowermost Deccan normal sequences at the base of Narmada section with the Cretaceous long superchron (34N, i.e., > 80 Ma), and suggested that volcanism lasted for more than 25 Ma. On the basis of palaeontological, palaeomagnetic and K-Ar results Courtillot et al. [7] disputed this correlation. Both Courtillot et al. [7] and Baksi [1] suggested that Deccan tectonic activity correlated with 29R (ca. 65 Ma), and that magmatic activity lasted for a relatively short period (ca. < 5 Ma). $^{40}Ar/^{39}Ar$ isotope ages reported for these important volcanic sequences [21–25] have been compiled by Vandamme et al. (Table 3 [19]). These are compared with the present results and the palaeomagnetic stratigraphic record of the Deccan traps in Fig. 7. Most units of the Deccan traps are reversely polarised, with isochron ages between

Table 1
Palaeomagnetic data for the central Kerala dykes, India

Site	Dyke trend	Lat.	Long.	N	N_s	D	I	k	α_{95}	λ'	ϕ'
* Kot-1	NNW	9.93	76.60	8	8	308	−48	66	6.9	26.8	307.2
* Kot-5	NNW	9.76	76.66	7	7	309	−62	117	5.6	19.9	293.5
* Kot-6	NNW	9.88	76.61	6	6	304	−60	241	4.9	18.0	297.7
Mean swarm vector						307	−57	107	12.0		
Mean VGP: 21.6, 299.4								*Palaeolatitude:* 31.8S			
* Kot-2	NW	9.96	76.58	7	7	328	−61	142	5.1	30.5	284.0
* Kot-3	NW	9.90	76.59	7	7	194	57	41	9.6	41.3	241.8
+ Kot-4	NW	9.91	76.55	6	6	242	44	40	10.6	20.4	198.9
* Kot-7	NW	9.88	76.60	5	5	163	66	226	5.1	29.9	269.6
* Kot-8	NW	9.95	76.54	6	6	153	66	50	9.6	26.7	276.5
* Kot-9	NW	9.37	76.77	7	7	149	56	37	10.1	36.0	287.6
@ Kot-10	NW	9.41	76.81	7	–	–	–	–	–	–	–
* Kot-11	NW	9.47	76.84	7	7	156	57	374	3.1	37.9	280.9
Mean swarm vector						163	61	58	10.1		
Palaeomagnetic pole: 34.6, 374.03								*Palaeolatitude:* 31.6S			

n = number of samples studied from the site; N_s = number of samples yielding stable vectors; D and I = the declination and inclination (in degrees), respectively, of the characteristic remanent magnetic vectors; k = precision parameter; α_{95} = radius of the 95% confidence cone about the mean direction; λ' and ϕ' = the latitude and longitude respectively of the VGP. * Site mean values included to calculate mean swarm vectors; + anomalous or a distinct direction that is not considered for the mean swarm vectors, @ site not considered for mean calculation, because it either failed to a yield stable end point or grouping.

69 and 65 Ma. A 69 Ma crystallisation age of the central Kerala dolerites and their reverse polarity coupled with available data on the Deccan indicates that these dykes were coeval with the significant volcanic episode which may correspond to the 31R chron. In addition, the geochemical and petrological characteristics of the central Kerala dolerites [6] suggest comparisons between the dykes and the Deccan trap sequences. These suggest that the dykes were probably feeders for Deccan volcanic units. This magmatic episode appears to have lasted for ca. 2–3 Ma (69–66 Ma), closely matching with the Cretaceous/Tertiary boundary.

The leucogabbro yielded a mean VGP of $\lambda = 21.6$, $\phi = 299.4$ ($\alpha_{95} = 12°$). The ages fall in the

Table 2
$^{40}Ar/^{39}Ar$ analytical data for incremental heating experiments on whole-rock dolerite samples from central Kerala, India

Release Temp. (°C)	$^{40}Ar/^{39}Ar$ *	$^{36}Ar/^{39}Ar$ *	$^{37}Ar/^{39}Ar$ [c]	% ^{39}Ar of total	% ^{40}Ar (Non-atmos.) [+]	% $^{36}Ar_{Ca}$	Apparent age (Ma) **
Monitor: MMhb-1 (519.4 ± 3.2 Ma, [11])							
Sample K2G: J = 0.010035							
450	665.28	2.21846	5.695	3.11	1.53	0.07	176.0 ± 83.2
500	266.92	0.87764	4.725	2.35	2.98	0.15	139.0 ± 41.2
550	179.69	0.59284	4.373	1.69	2.70	0.20	86.0 ± 30.6
580	96.20	0.30284	4.492	4.15	7.34	0.40	123.9 ± 12.1
610	46.58	0.13437	4.747	3.18	15.56	0.96	127.1 ± 7.0
640	31.91	0.08994	4.711	4.17	17.88	1.42	100.8 ± 6.3
700	17.73	0.04562	6.909	6.24	27.07	4.14	85.2 ± 4.2
780	15.79	0.04288	15.242	10.22	27.48	9.67	77.6 ± 2.6
820	19.38	0.05503	15.416	11.28	22.47	7.62	77.9 ± 4.3
860	23.23	0.06561	12.543	8.51	20.86	5.20	86.3 ± 4.1
900	20.36	0.05557	9.905	6.48	23.23	4.85	84.2 ± 4.6
950	17.90	0.04673	8.498	8.12	26.63	4.95	84.7 ± 4.5
1000	11.50	0.02700	6.671	11.92	35.21	6.72	72.1 ± 3.2
1050	10.34	0.02621	15.315	9.04	36.93	15.89	68.5 ± 3.3
Fusion	38.10	0.12384	67.358	9.56	18.1	14.79	126.1 ± 6.4
Total	52.39	0.16403	15.642	100.00	23.45	7.03	92.1 ± 13.2
Total without 450–640°C and fusion				71.79			78.7 ± 4.4
Sample K3B: J = 0.009865							
450	296.52	0.98066	6.978	2.10	2.46	0.19	125.8 ± 39.3
500	77.90	0.24525	4.945	2.14	7.48	0.55	101.1 ± 10.2
560	53.66	0.14665	4.917	1.09	19.97	0.91	181.8 ± 16.4
610	39.82	0.11529	4.769	3.11	15.39	1.13	106.2 ± 7.0
660	35.72	0.10100	4.709	4.14	17.49	1.27	108.2 ± 7.4
710	20.46	0.05339	7.521	5.57	25.82	3.83	92.1 ± 2.6
760	23.79	0.07012	13.127	9.88	17.29	5.09	72.3 ± 2.2
830	27.46	0.08204	12.878	15.62	15.46	4.27	74.6 ± 1.5
890	28.19	0.08349	8.977	14.72	15.02	2.92	74.3 ± 1.6
950	11.98	0.02887	5.327	15.81	32.31	5.02	67.8 ± 2.2
1010	7.73	0.01392	5.437	14.29	52.37	10.63	70.9 ± 1.3
1070	11.84	0.03258	23.191	9.11	34.36	19.36	72.0 ± 2.0
Fusion	35.93	0.13152	128.897	2.03	20.66	26.66	139.4 ± 8.7
Total	28.07	0.08290	12.565	100.00	25.63	6.65	80.2 ± 4.0
Total without 450–710°C and fusion				79.42			71.9 ± 1.5

Table 2 (continued)

Release Temp. (°C)	$^{40}Ar/^{39}Ar$ *	$^{36}Ar/^{39}Ar$ *	$^{37}Ar/^{39}Ar$ c	% ^{39}Ar of total	% ^{40}Ar (Non-atmos.) +	% $^{36}Ar_{Ca}$	Apparent age (Ma) **
Sample K3C: J = 0.009551							
450	203.40	0.67290	7.045	3.51	2.73	0.28	93.8 ± 61.2
510	52.56	0.16387	6.422	1.84	8.84	1.07	78.6 ± 18.3
570	35.25	0.09915	6.135	1.77	18.26	1.68	108.0 ± 8.2
620	27.66	0.07763	5.954	2.56	18.78	2.09	87.7 ± 8.6
680	19.65	0.04581	7.178	9.26	34.03	4.26	112.2 ± 2.0
740	14.69	0.03216	11.697	6.29	41.68	9.89	103.3 ± 3.3
790	15.32	0.03703	14.772	10.94	36.30	10.85	94.2 ± 2.5
840	26.90	0.06731	14.145	15.23	30.28	5.72	136.3 ± 2.8
890	22.69	0.05381	9.177	13.77	33.16	4.64	125.9 ± 1.6
940	13.98	0.02899	5.577	12.90	41.88	5.23	98.5 ± 1.6
990	9.42	0.01644	6.421	10.22	53.84	10.62	85.7 ± 1.8
1040	8.48	0.01568	15.922	7.44	60.42	27.63	87.1 ± 2.0
Fusion	28.29	0.10105	108.596	4.27	25.28	29.23	127.7 ± 3.6
Total	25.70	0.06925	14.258	100.00	36.46	8.89	107.6 ± 4.2
Sample K4D: J = 0.009045							
450	73.57	0.14691	5.776	13.40	41.61	1.07	442.4 ± 6.2
510	28.85	0.01407	5.291	25.89	87.04	10.23	370.4 ± 2.3
570	25.88	0.01963	5.207	3.97	79.18	7.22	370.6 ± 2.4
630	23.92	0.02277	7.663	6.66	74.42	9.15	270.6 ± 2.5
680	23.83	0.02896	19.175	5.00	70.54	18.01	258.3 ± 2.6
730	26.72	0.04007	18.077	7.05	61.10	12.27	251.2 ± 2.9
780	28.16	0.03493	14.491	9.25	67.46	11.28	288.5 ± 3.3
830	32.09	0.05633	15.800	5.53	52.06	7.63	256.2 ± 3.4
880	26.54	0.02722	9.513	5.85	72.54	9.51	291.2 ± 4.2
930	23.07	0.02661	8.193	5.17	67.33	8.38	228.6 ± 3.3
980	21.79	0.01369	9.063	4.32	84.75	18.00	280.1 ± 4.1
1030	23.07	0.01863	18.056	3.89	82.40	26.36	289.2 ± 4.8
Fusion	27.94	0.01224	40.177	4.03	98.59	89.29	410.7 ± 8.2
Total	33.06	0.04102	11.000	100.00	71.72	13.31	325.6 ± 4.0

* measured; c corrected for post-irradiation decay of ^{37}Ar (35.1 day half-life). + $[^{40}Ar_{tot.} - (^{36}Ar_{atmos.})(295.5)]/^{40}Ar_{tot.}$; ** calculated using correction factors of Dalrymple et al. [16] (2σ intralaboratory errors).

windows of 70–90 Ma (80 Ma mean) and 80–100 Ma (90 Ma mean) [18]. Based on the 80 Ma and 90 Ma mean poles on the APWP and their weighted ages, an 84.9 ± 7.6 Ma palaeomagnetic age is interpreted for the leucogabbro dyke. This is in agreement with the 82 ± 3 Ma K-Ar age.

Table 3
$^{36}Ar/^{40}Ar$ vs. $^{39}Ar/^{40}Ar$ isotope correlations from incremental heating experiments on whole-rock dolerite samples from central Kerala, India

Sample	Isotope correlation age (Ma) *	$^{40}Ar/^{39}Ar$ intercept **	MSWD	Increments included (°C)	Percent of total ^{39}Ar
K2G	59.7 ± 5.3	321.7 ± 7.3	0.52	700–1050	71.79
K3B	69.4 ± 1.1	298.8 ± 2.3	0.80	760–1070	79.42
K3C	69.9 ± 2.0	357.5 ± 3.0	0.62	680–1040	86.05

* calculated using the inverse abscissa intercept ($^{40}Ar/^{39}Ar$ ratio) in the age equation; ** inverse ordinate intercept.

The normal polarity and the assigned age of this leucogabbro may correlate with emplacement during chron 34N. These ages contrast with the $^{40}Ar/^{39}Ar$ age results reported for the Deccan traps (Fig. 7). The leucogabbro has the geochemical characteristics of T-OIB, distinguishing it from the main Deccan trap rocks and the T-MORB-type central Kerala dolerites. Therefore, the leucogabbro is probably related to a distinctly different magmatic event. One sample (IG-02) from the Deccan region [25] yielded 82–83 Ma total fusion and isochron ages. Kaneoka [25] suggested that this relatively old volcanic activity may have corresponded to an episode that was precur-

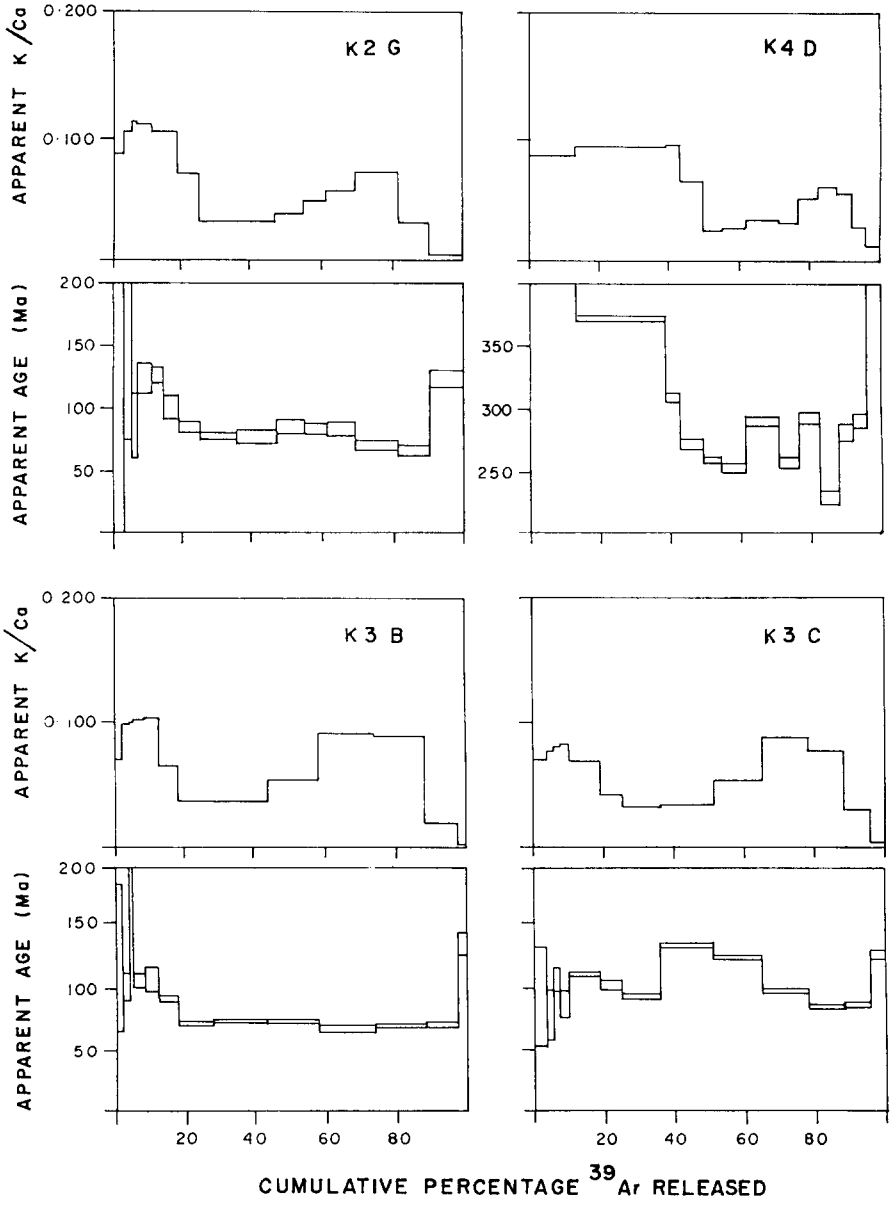


Fig. 5. $^{40}Ar/^{39}Ar$ age and apparent K/Ca spectra of whole-rock dolerite dyke samples from central Kerala.

sory ot the main volcanic activity. If this is correct, the sample must be within a tectonically disturbed area because such an old age is not stratigraphically possible in view of the 67–68 Ma ages reported from the stratigraphically lower Jawahar traps [22]. However, the older age (ca. 81 Ma) and the T-OIB characteristics of the leucogabbro and the T-MORB characteristics together with the younger ages (69–60 Ma) recorded for the central Kerala dolerite dykes are consistent with geotectonic models which include an earlier intraplate plume leading to a late asthenospheric decompressional tectonic setting prior to opening of the northwestern Indian Ocean [6].

7. Central Kerala dykes in relation to the mantle plume and the opening of northwestern Indian Ocean

The 81 Ma age and normal polarity of the leucogabbro of central Kerala correspond to 34N.

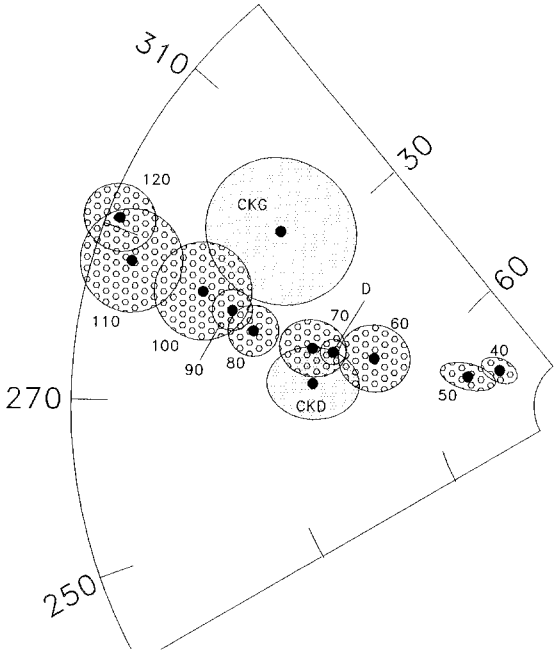


Fig. 6. Comparison of the palaeomagnetic poles of the central Kerala dykes with the synthetic APWP of India [18]. *CKD* and $CKG = \alpha_{95}$ ellipses of mean poles of, respectively, dolerites and gabbros of central Kerala. Mean Deccan trap pole (*D*) and its α_{95} ellipse [19] are also shown for comparison.

This magmatism appears to have been coeval with felsic dyke intrusions at 81 ± 2 Ma along the western coast of India [30]. It is also consistent with the ca. 84 Ma basalt eruptions on the eastern flank of the Amirante ridge [31]. The leucogabbros display T-OIB geochemical characteristics [6], suggesting an origin within an intraplate thermal plume. Magnetic anomaly data [32] from the western Indian Ocean show that the Mascarene–Seychelles–India tectonic element remained as a compact landmass which began drifting from Madagascar–Africa at ca. 80 Ma. The ca. 81 Ma age of the T-OIB-type alkaline dyke was possibly not related to the spreading because alkaline magmatism generally precedes tholeiitic volcanism and initiation of spreading [33]. Thus the early leucogabbros and other magmatic events at c. 81 Ma in western India could record a precursory igneous episode triggered from a thermal plume. The tectonic situation along western India changed dramatically after an interval of ca. 1O Ma, when the region was affected by extensive volcanism in the form of the Deccan traps and several dyke intrusions similar to the central Kerala dolerites, at ca. 69–65 Ma. Magma production rates were high, suggesting relatively greater rates of crustal extension and correspondingly faster rates of asthenospheric upwelling. This could have resulted in the eruption of dominantly T-MORB-type rocks. Cessation of this peak magmatic episode coincided with the Cretaceous/Tertiary boundary.

It is likely that an intraplate plume tectonic setting existed at ca. 80 Ma, and subsequently led to eruption of a large volume of T-MORB-type magma (Deccan basalts and related dykes). The recent hypothesis [4,34] of plume initiation is governed by the argument that the main Deccan (flood basalt) event was not preceded by large degrees of extension or alkaline volcanism (typical of volcanism associated with extension). However, this may be an artifact of the lack of systematic isotope dating of the Deccan alkaline rocks [33]. Therefore, our ca. 81 Ma age for the leucogabbro dyke, which is of an alkaline character, is significant because it does not support this tectonic scenario. Alternatively, our results suggest clear evidence for extension and the existence of

a plume in western India prior to the large Deccan eruptions. The Deccan and related rocks in western India may have formed by decompression melting of a hot plume head beneath a region already undergoing extension in response to external plate-driving mechanisms. Using planation surfaces [35] and dome-flank drainage patterns [36] a similar model has been suggested [33] based on uplift and extension inferred in western India prior to Deccan activity. Thus, the Deccan and related western Indian volcanism is likely to reflect plume incubation for a period of ca. 10–15 Ma resulting in lithospheric thinning and decompressional melting.

The time span between ca. 80 Ma and ca. 60 Ma was marked by thinning of continental crust followed by ocean-floor spreading between the Seychelles–Mascarene and India. Seafloor anomaly data record that spreading between the Africa–Madagascar and Mascarene–Seychelles–India continents terminated at anomaly 28 (66–65 Ma), and shifted northward to initiate formation of the Arabian Sea prior to anomaly 27 (ca. 63–64 Ma) [32]. A 63 Ma age reported for the

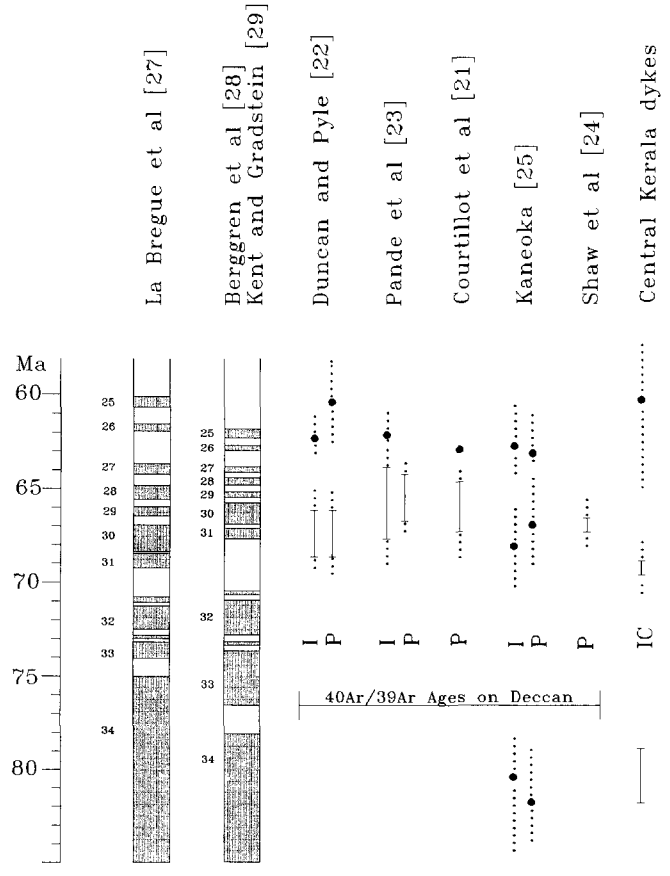


Fig. 7. Central Kerala dyke age data compared with the data on Deccan traps and the magnetic polarity time scale. The numerical ages for the polarity scale on the left are after Mankinen and Dalrymple [26]. *P* = Plateau ages; *I* = isochron ages; *IC* = isotope correlation ages. Two lower plateau ages on MAP-056 and CAT-034 [22] have not been included because of large differences between the plateau and isochron ages and lower $^{40}Ar/^{36}Ar$ intercepts. The range of absolute ages are marked as continuous lines and error limits are extended as dotted lines. Note that a good number of ages covers a major portion of 31R, and palaeomagnetically the Deccan lavas as well as the central Kerala dolerite dykes are predominantly reversely magnetised. Details are discussed in the text.

basaltic magmatism on the Seychelles [37] may also correspond to this major geodynamic activity. The younger ages (63.6 ± 0.5 Ma) reported for a dyke southwest of Indore in the Deccan region [21] and our younger age (60 Ma) with normal polarity could have been associated with the waning stages of the flood volcanic episode in India, and could have coincided with transformation of a rift setting into ocean-floor spreading and opening of the Arabian Sea.

8. Conclusions

Two distinct dyke systems occur in central Kerala, between 9° and 10°N along the western coast of India. Predominant are the NW-trending dolerites. These dykes most commonly possess a reverse palaeomagnetic polarity, like most of the Deccan trap units, and $^{36}Ar/^{40}Ar$ vs. $^{39}Ar/^{40}Ar$ isotope correlations have yielded an age of ca. 69 Ma. Based on these comparisons, it has been suggested that the central Kerala dolerites may correspond to 31R and could be feeders of the Deccan traps. The NNW-trending leucogabbro dated at ca. 81 Ma possesses a normal palaeomagnetic vector and corresponda to the 34N superchron. The leucogabbro might represent a precursory magmatic event from a thermal plume. The data support a geodynamic model comprising an early intraplate plume tectonic setting (ca. 80 Ma) gradually turning into a late asthenospheric decompressional tectonic setting (ca. 70–65 Ma) which finally lead to the rifting of the Seychelles from India and opening of the Arabian Sea between the Seychelles and India.

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